1. Even or Odd Function

Code:

```
def is_even(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

num = int(input("Enter a number: "))
print(is_even(num))
```

Output:

```
Enter a number: 7
Odd
```

2. Largest of Two Numbers

Code:

```
def largest(a, b):
    if a > b:
        return a
    else:
        return b

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Largest number is:", largest(a, b))
```

Output:

```
Enter first number: 25
Enter second number: 40
Largest number is: 40
```

3. Print Numbers

Code:

```
def print_numbers(n):
    for i in range(1, n + 1):
        print(i)

n = int(input("Enter a number: "))
print_numbers(n)
```

Output:

```
Enter a number: 5
1
2
3
4
5
```

4. Sum of Numbers

Code:

```
def sum_n(n):
    total = 0
```

```
    for i in range(1, n + 1):
        total += i
    return total

n = int(input("Enter a number: "))
print("Sum is:", sum_n(n))
```

Output:

```
Enter a number: 5
Sum is: 15
```

5. Multiplication Table

Code:

```
def table(n):
    for i in range(1, 11):
        print(n, "x", i, "=", n * i)

n = int(input("Enter a number: "))
table(n)
```

Output:

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

6. Count Even Numbers

Code:

```
def count_even(numbers):
    count = 0
    for num in numbers:
        if num % 2 == 0:
            count += 1
    return count

numbers = [1, 2, 3, 4, 6]
print("Count of even numbers:", count_even(numbers))
```

Output:

```
Count of even numbers: 3
```

7. Find Maximum Without Using max()

Code:

```
def find_max(numbers):
    largest = numbers[0]
    for num in numbers:
        if num > largest:
            largest = num
```

```
    return largest

numbers = [10, 25, 5, 40, 15]
print("Largest number:", find_max(numbers))
```

Output:

```
Largest number: 40
```

8. FizzBuzz

Code:

```
def fizzbuzz():
    for i in range(1, 101):
        if i % 3 == 0 and i % 5 == 0:
            print("FizzBuzz")
        elif i % 3 == 0:
            print("Fizz")
        elif i % 5 == 0:
            print("Buzz")
        else:
            print(i)

fizzbuzz()
```

Output:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
...
The output continues up to 100.
```

9. Palindrome Checker

Code:

```
def palindrome(s):
    if s == s[::-1]:
        return True
    else:
        return False

text = input("Enter a string: ")
```

```
print(palindrome(text))
```

Output:

```
Enter a string: madam
True
```

10. Count Vowels

Code:

```
def count_vowels(s):
    count = 0
    for ch in s:
        if ch.lower() in "aeiou":
            count += 1
    return count

text = input("Enter a string: ")
print("Number of vowels:", count_vowels(text))
```

Output:

```
Enter a string: Hello World
Number of vowels: 3
```

11. Second Largest Number Without Using sort()

Code:

```
def second_largest(numbers):
    largest = float('-inf')
    second = float('-inf')

    for num in numbers:
        if num > largest:
            second = largest
            largest = num
        elif largest > num > second:
            second = num

    if second == float('-inf'):
        return "Second largest does not exist"

    return second

numbers = [10, 20, 40, 30, 50]
print("Second largest number:", second_largest(numbers))
```

Output:

```
Second largest number: 40
```

12. Guessing Game

Code:

```
import random

def guessing_game():
    number = random.randint(1, 10)

    while True:
```

```
        guess = int(input("Guess a number between 1 and 10: "))

        if guess == number:
            print("Congratulations! Correct guess.")
            break
        elif guess < number:
            print("Too low! Try again.")
        else:
            print("Too high! Try again.")

guessing_game()
```

Output:

```
Guess a number between 1 and 10: 3
Too low! Try again.
Guess a number between 1 and 10: 8
Too high! Try again.
Guess a number between 1 and 10: 6
Congratulations! Correct guess.
```

13. Grade Calculator

Code:

```
def grade(marks):
    if marks >= 90:
        return "A"
    elif marks >= 80:
        return "B"
    elif marks >= 70:
        return "C"
    elif marks >= 60:
        return "D"
    else:
        return "F"

marks = int(input("Enter marks: "))
print("Grade:", grade(marks))
```

Output:

```
Enter marks: 85
Grade: B
```